Aim: Implementation to gather information from any PC's connected to the LAN using who.is, port scanners, network scanning, Angry IP scanners etc.

Objective: To know how to gather information about the networks by using different n/w reconnaissance tools.

Requirements: Laptop, who.is, n map, angry ip scanner

1. Who.is

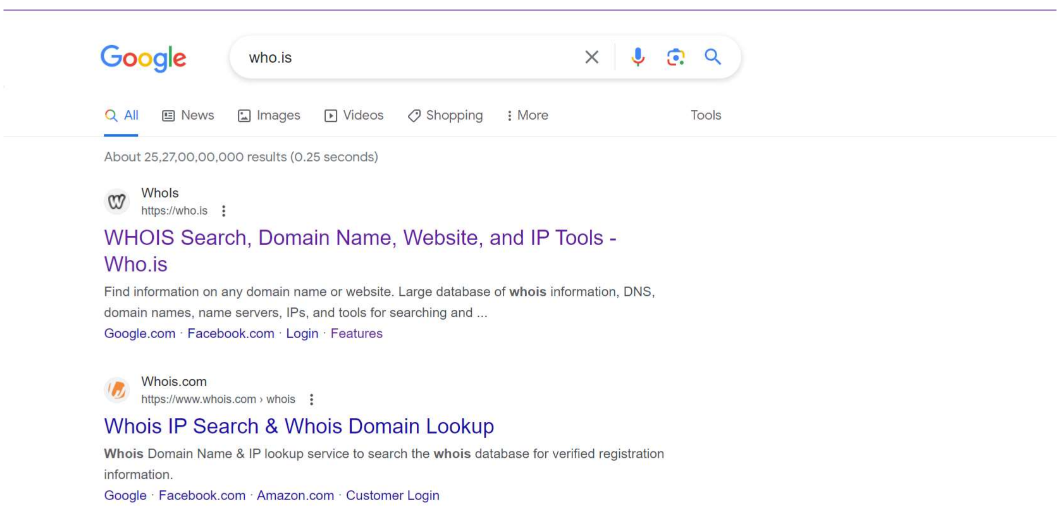
- Theory

whois search for an object in a WHOIS database. WHOIS is a query and response protocol that is widely used for querying databases that store the registered users of an Internet resource, such as a domain name or an IP address block but is also used for a wider range of other information. Most modern versions of whops try to guess the right server to ask for the specified object. If no guess can be made, whops will connect to whops.networksolutions.com for NIC handles or whops.arin.net for IPv4 addresses and network names.
Examples: Obtaining the domain WHOIS record for computer solutions.com WHOIS record by IP querying Querying WHOIS in google search engine

To use the WHO.IS lookup tool, just enter the domain name whose information you'd like to view into the search field on the WHOIS main page. You can retrieve key data about a domain in this way, including availability, **domain owner lookup**, and creation and expiration details. If you own multiple domains of your own, it can be helpful to download exportable lists from the tool to analyze large amounts of domains data.

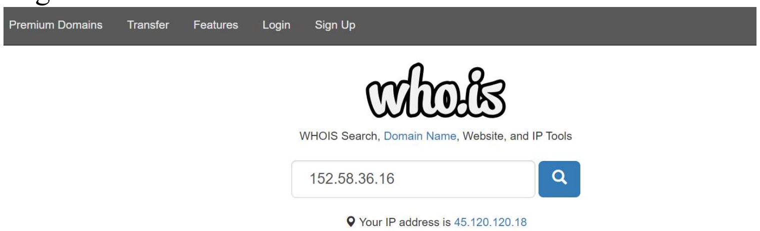
- **Implementation**

Step 1: search who.is in web browser



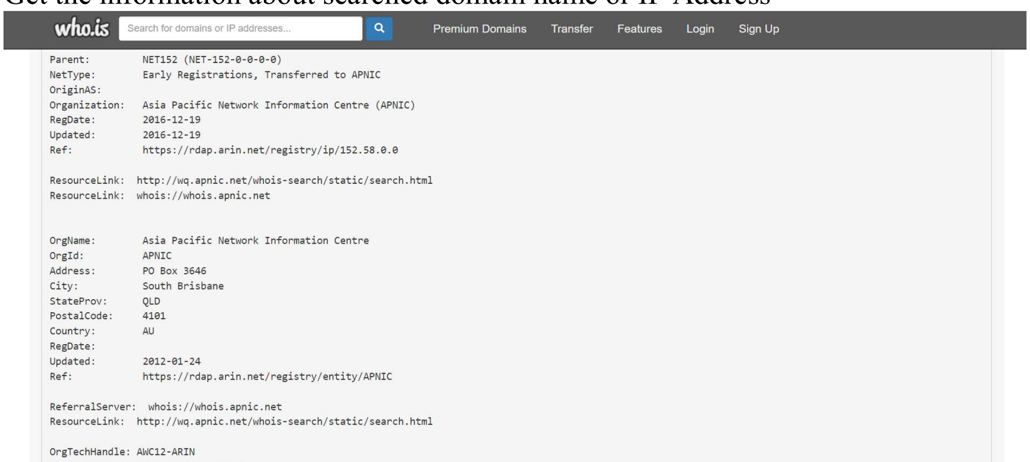
Step 2: open the who.is tool

Step 3: Enter the Domain name or IP Address of which devices' information you want to gather



- **Observation:**

Get the information about searched domain name or IP Address



2. Port Scanners:

Nmap is convenient during penetration testing of networked systems. Nmap provides the network details, and also helps to determine the security flaws present in the system. Nmap is **platform-independent** and runs on popular **operating systems** such as **Linux, Windows** and **Mac**.

Nmap is a useful tool for network scanning and auditing purposes.

- It can search for hosts connected to the Network.
- It can search for free **ports** on the target host.
- It detects all services running on the host with the help of **operating system**.

- It also detects any **flaws** or **potential vulnerabilities** in networked systems.

It is effortless to work with the Nmap. With the release of a new graphical user interface called **ZenMap User**, it performs many tasks such as saving and comparing scan results, scanning the results in a database, and visualize the network system topology graphically, etc.

- **Advantages of Nmap:**

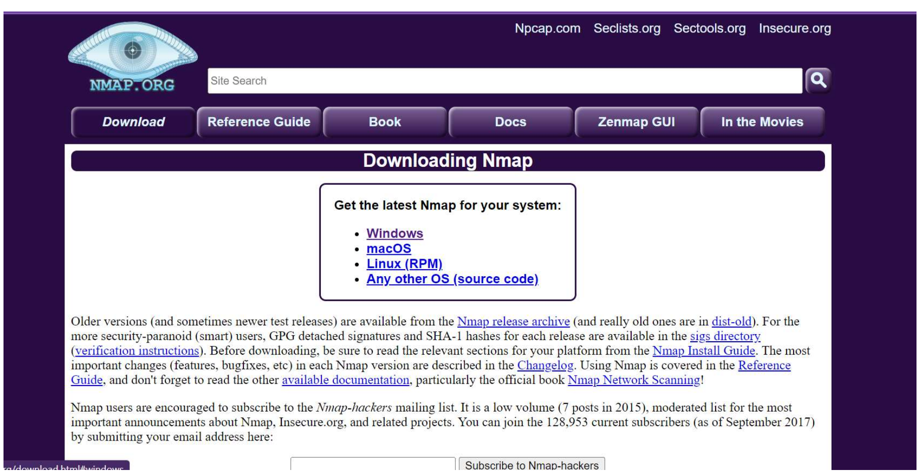
Nmap has a lot of advantages that make it different from other network scanning tools. Nmap is open-source and **free** to use.

Some other advantages are listed below.

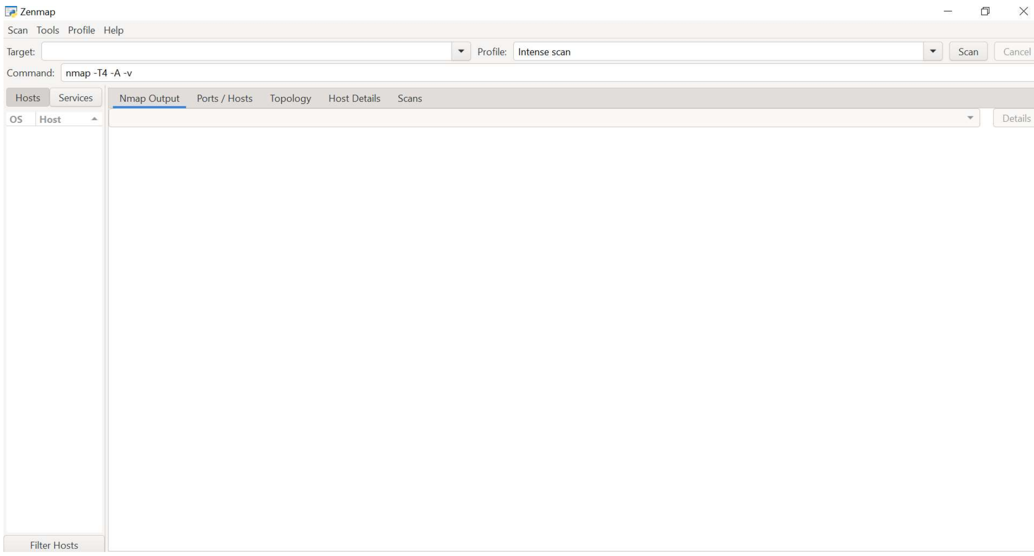
- It is used for auditing network systems as it can detect new servers.
- It will search for **subdomain** and Domain Name Systedm
- With the help of Nmap Scripting Engine (**NSE**), interaction can be made with the target host.
- It determines the nature of the service in the host and performs whether the host is a mail service or a web server

- Port scanning using Nmap:
- **Implementation:**

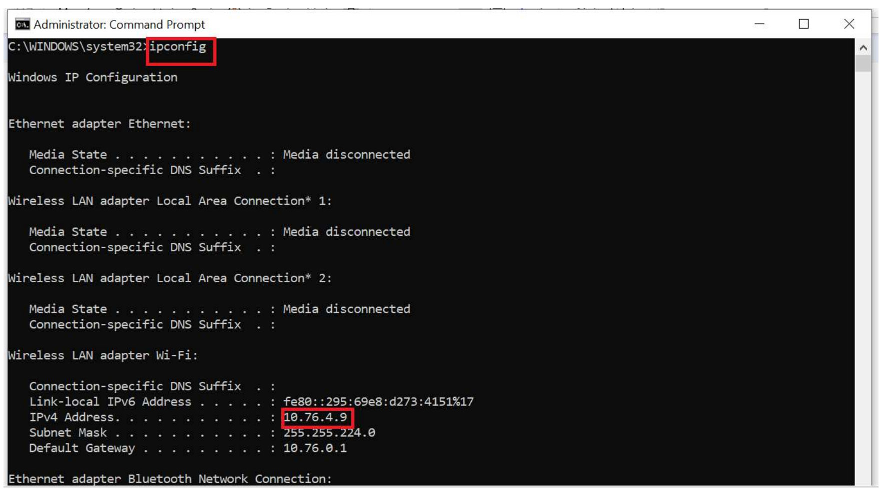
Step 1: Install Nmap in Windows



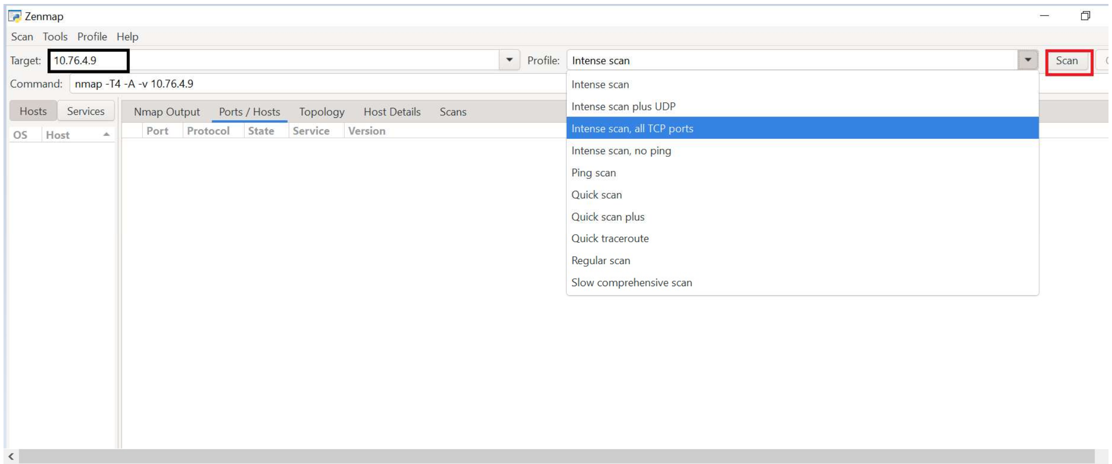
Step 2: Open Nmap



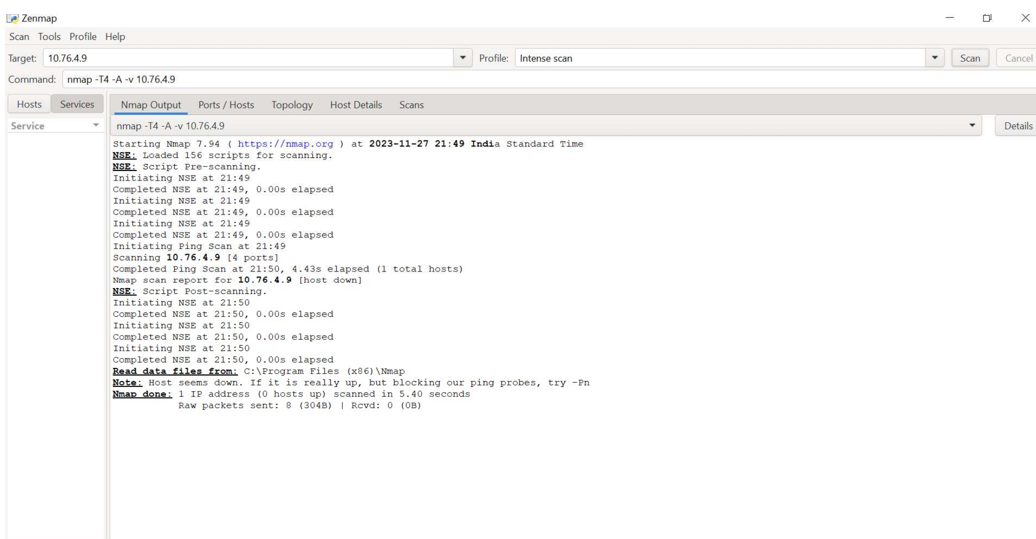
Step 3: Open command prompt and give command ipconfig to see local network



Step 4: paste ip address in target select scan type and press scan button

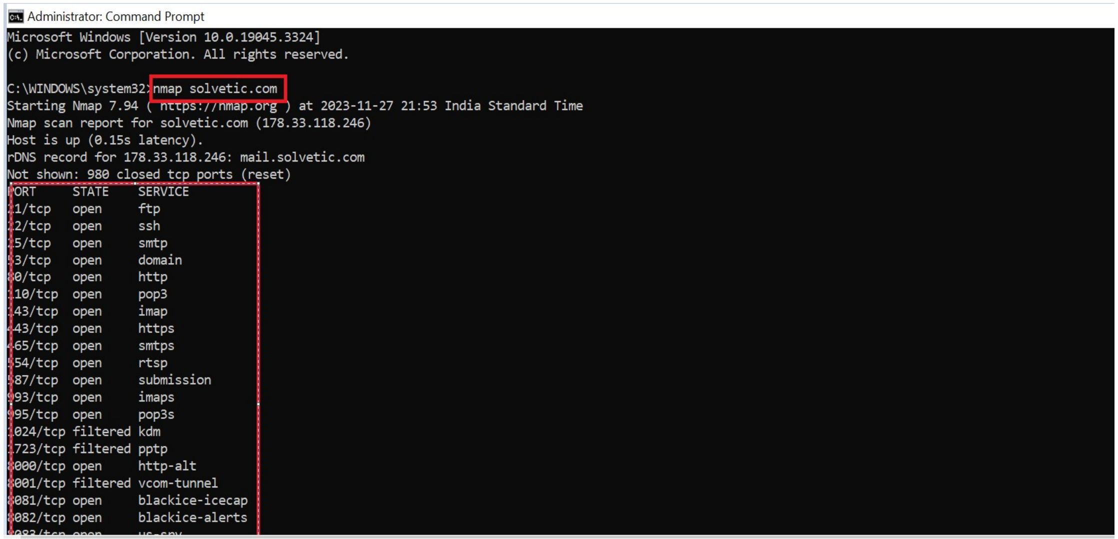


Nmap Output:



Step 5: Open Command prompt Run as administrator

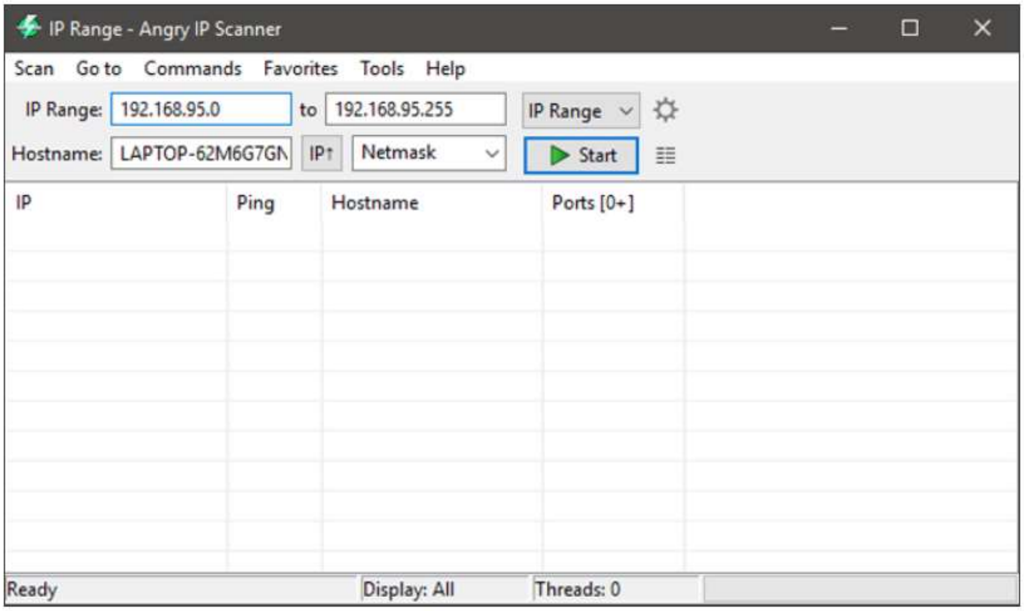
To see the open ports run:



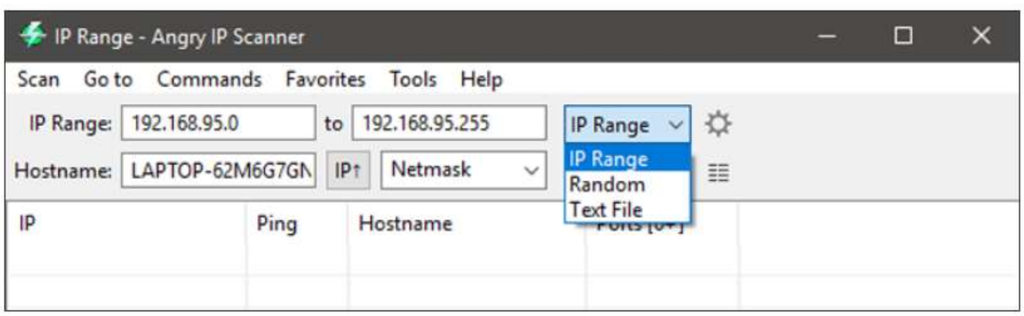
3. Angry IP Scanner

- **Implementation:**

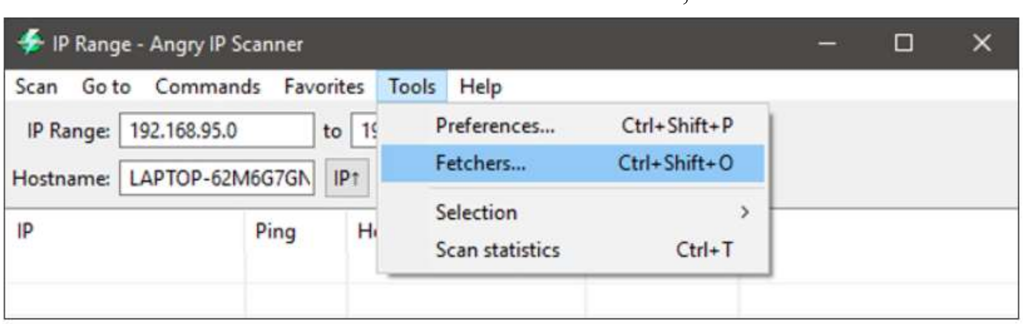
Once installed, open the application by searching for it in the Start Menu. As you can see, the home screen of the application is pretty simple and straightforward. By default, Angry IP scanner will enter your local IP address range and your computer name as the hostname.



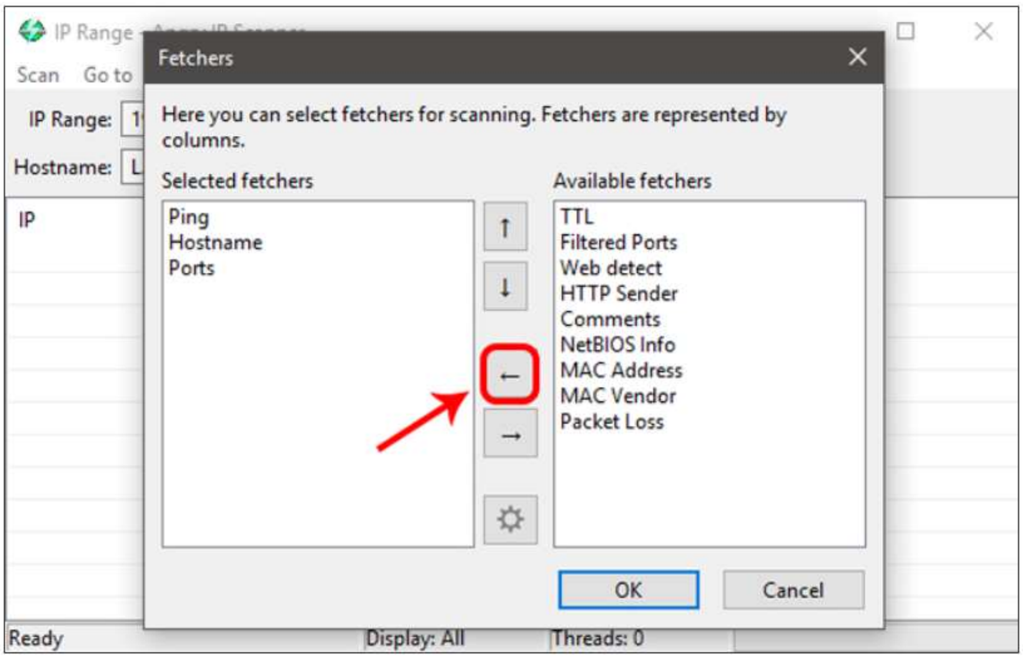
The good thing about Angry IP Scanner is that it lets you scan IP addresses in three different ways. They are, the range you specified, a random IP address or a list of IP addresses from a text file. You can easily select the scan mode from the drop-down menu next to the IP address field.



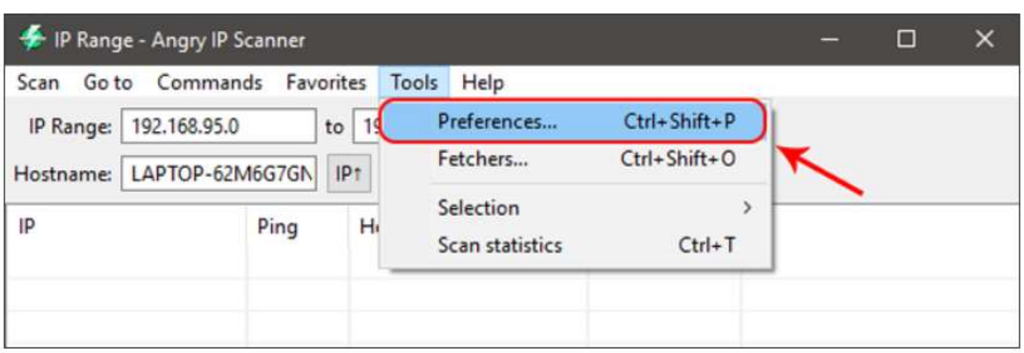
As you can see from the above image, the Angry IP Scanner will only include default fetchers like Ping, Hostname, and Ports. However, you can add more fetchers to get and see more information about an IP address. To do that, select “Tools > Fetchers.”



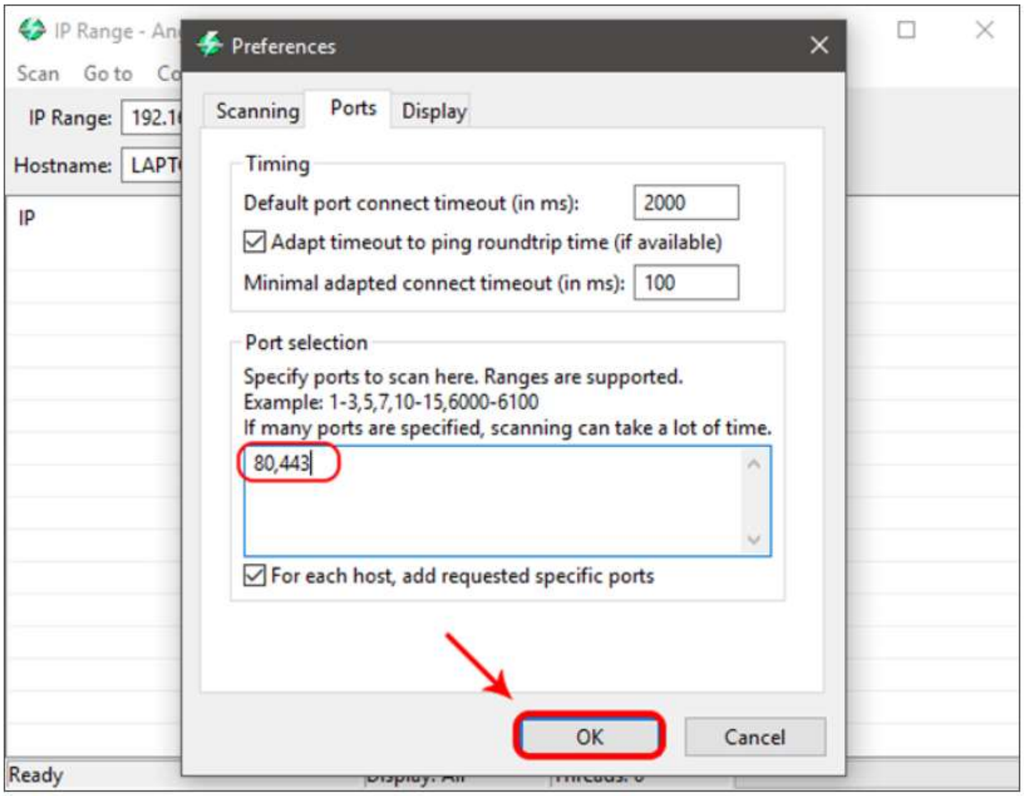
In this window, you will see all the current fetchers on the left pane and all the available fetchers in the right pane. To add a fetcher, select the fetcher on the right pane and then click on the button that looks like "Less than" sign. In my case, I've added new fetchers like MAC address, NetBIOS info, filtered ports, and Web detectors.



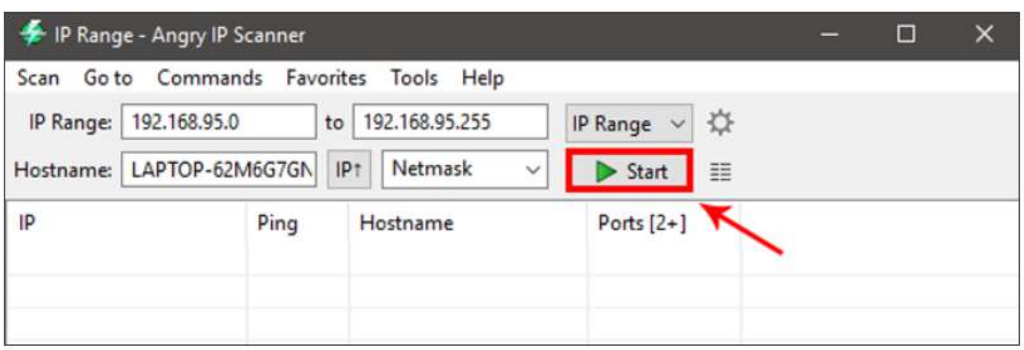
Moreover, Angry IP Scanner will only tell whether the ports are open or not. It will not list the individual ports that are open. So, if you want to do a port scan, then you need to configure the application. To do that, simply navigate to "Tools" and then select the option "Preferences."



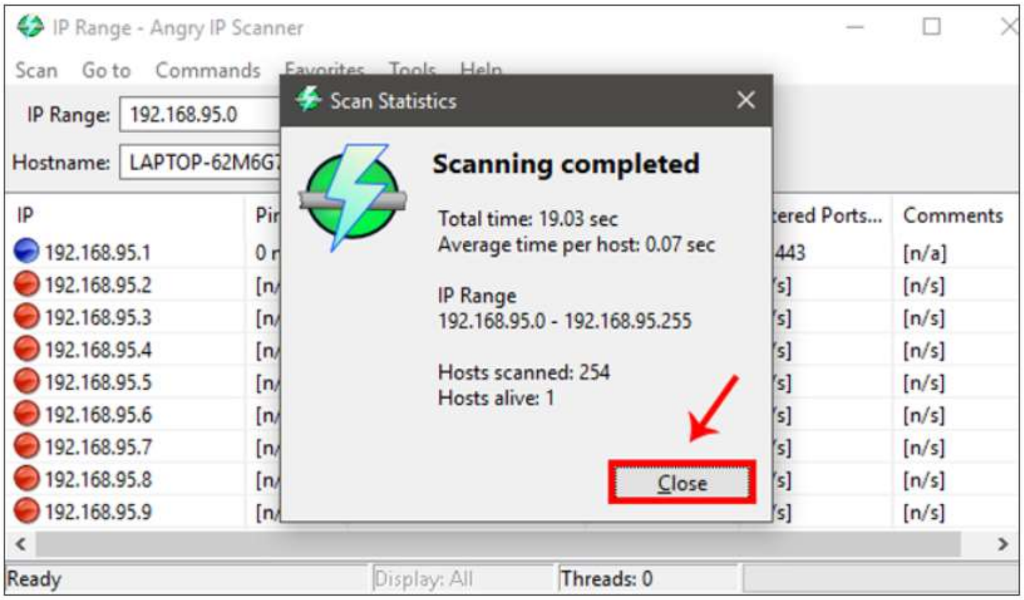
Here, navigate to the "Ports" tab and enter the ports you want to scan in the "Ports Selection" field. As you can see, I'm scanning for port 80 and port 443. If you want to scan a range of ports, then enter the port range like "1-1000." After you have entered the port number, click OK to save the changes.



Once you are done configuring the Angry IP Scanner, you can continue to scan. To start off, set the scan mode to "IP Range," enter the IP address range in the "IP address" fields and then click on the button "Start." For instance, I've entered an IP range that is known to have live devices connected to it.



Depending on the number of addresses in the range, it may take some time to complete. Once completed, the application will show you a summary of the scan. The summary includes the number of hosts that are alive and the number of hosts that have open ports. Just click on the button "Close" to continue.



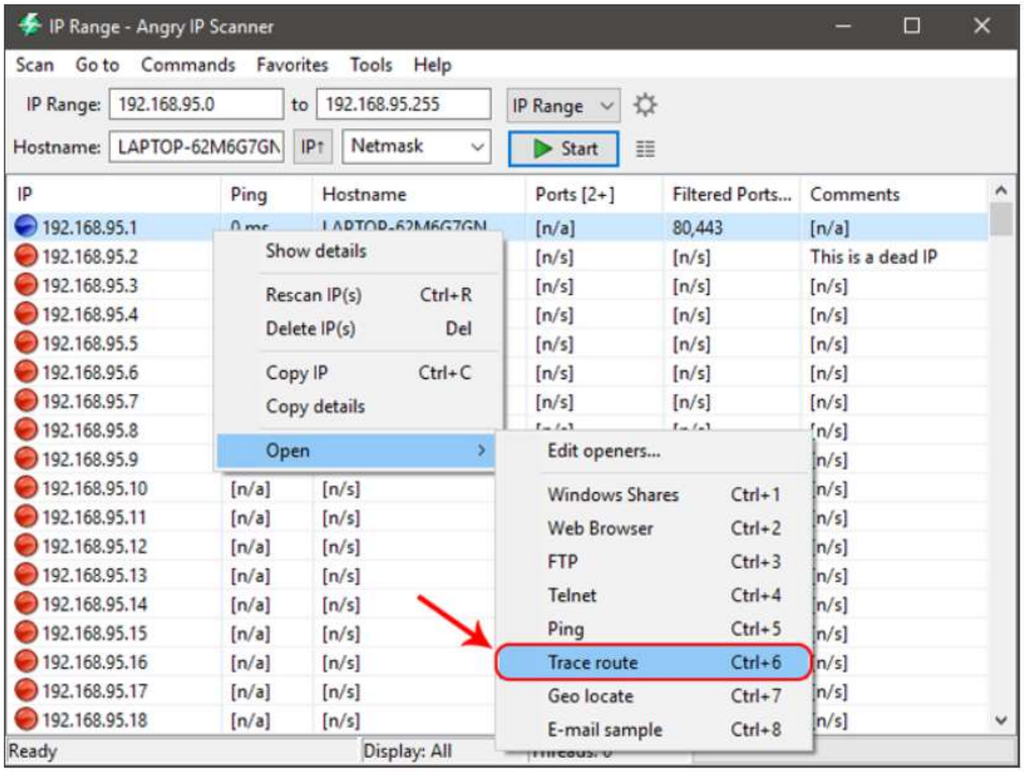
Once you close the summary window, you will see the list of all the IP addresses. You can also see additional details in different "fetcher" columns. In case you are wondering, here's what the colored dots next to each IP address mean.

Red: The IP address is inactive, dead or there is no device connected to this IP address.

Blue: The IP address is either active or busy and not responding to the requests sent by Angry IP Scanner. This usually will be your own IP Address.

Green: The IP address is active, and the device connected to it is responding to the requests made by Angry IP Scanner. There may also be open ports.

Apart from copying the details of an IP address, you can also perform a range of different activities on the entries. You can open an IP address in the web browser, do an FTP, trace routing, etc. For instance, if you want to traceroute an IP address, simply right-click on the target IP address. After that, select the option Open and click on Traceroute.



Once you are done scanning an IP address or the IP address range, you can save the scan results. To do that, select the option Scan from the menu bar.